Supplementary Material for "TAB-PO: Preference Optimization with a Token-Level Adaptive Barrier for Token-Critical Structured Generation"

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Appendix A. Mathematical Intuition for Gradient Dilution and Preferred-Token Erosion

This appendix gives the mathematical intuition behind two failure modes that can arise when DPO-style preference optimization is applied to low-separation structured outputs: gradient dilution and preferred-token erosion. The main paper reports empirical mechanism-level diagnostics for TAB-PO and DPO-family variants in Section 7.1; here we focus only on why these failure modes arise under DPO.

Given a preference triple, let Y^+ and Y^- denote the preferred and rejected structured record sets for the same input, where each record has the form $y_i = (\ell_i, \Pi_i, R_i)$. Let x denote the input prompt. We serialize the two structured outputs as completion strings Y_s^+ and Y_s^-. Let $u^+ = (u_1^+, \ldots, u_{T^+}^+)$ and $u^- = (u_1^-, \ldots, u_{T^-}^-)$ denote the completion-token subsequences corresponding

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to Y_s^+ and Y_s^-, respectively. The reference model is the supervised model p_{SFT}.

For a single preference pair, standard DPO minimizes

$$\mathcal{L}_{\mathrm{DPO}}(\theta) = -\log \sigma\left(\beta\left[\log \frac{p_\theta(Y_s^+ \mid x)}{p_{\mathrm{SFT}}(Y_s^+ \mid x)} - \log \frac{p_\theta(Y_s^- \mid x)}{p_{\mathrm{SFT}}(Y_s^- \mid x)}\right]\right), \tag{A1}$$

where $\beta > 0$ controls the preference strength. Define the reference-adjusted preference margin

$$\Delta_\theta(x, Y_s^+, Y_s^-) = \log \frac{p_\theta(Y_s^+ \mid x)}{p_{\mathrm{SFT}}(Y_s^+ \mid x)} - \log \frac{p_\theta(Y_s^- \mid x)}{p_{\mathrm{SFT}}(Y_s^- \mid x)}. \tag{A2}$$

The gradient of the DPO loss is

$$\nabla_\theta \mathcal{L}_{\mathrm{DPO}} = -\beta\sigma(-\beta\Delta_\theta)\Big[\nabla_\theta \log p_\theta(Y_s^+ \mid x) - \nabla_\theta \log p_\theta(Y_s^- \mid x)\Big]. \tag{A3}$$

Using the autoregressive factorization

$$\log p_\theta(Y_s^\pm \mid x) = \sum_{t=1}^{T^\pm} \log p_\theta(u_t^\pm \mid x, u_{<t}^\pm),$$

we obtain

$$\begin{aligned}\nabla_\theta \mathcal{L}_{\mathrm{DPO}} = -\beta\sigma(-\beta\Delta_\theta)\Bigg[&\sum_{t=1}^{T^+} \nabla_\theta \log p_\theta\big(u_t^+ \mid x, u_{<t}^+\big) \\ &- \sum_{t=1}^{T^-} \nabla_\theta \log p_\theta\big(u_t^- \mid x, u_{<t}^-\big)\Bigg].\end{aligned} \tag{A4}$$

Let t_0 denote the first completion-token position at which u^+ and u^- differ, or the first position at which one completion terminates. For $t < t_0$, the preferred and rejected completions contain identical tokens under identical autoregressive prefixes, so their gradient terms cancel. The remaining DPO update is therefore determined by tokens at and after the first divergence. The

number of canceled pre-divergence tokens is task- and serialization-dependent. When these shared preferred tokens are under-confident under the SFT initialization, standard DPO provides no direct restoration signal because their preferred and rejected gradient contributions cancel exactly. Thus, SFT uncertainty can remain uncorrected even before the first structured divergence. TAB-PO addresses this gap by applying confidence-gated restoration to under-confident preferred tokens, including tokens whose DPO gradient contribution vanishes due to pre-divergence cancellation.

We group these remaining positions into three categories. Critical tokens are tokens that directly determine structured prediction correctness, including ontology labels, grounded evidence spans, relation labels, relation arguments, sentence identifiers, and coreference links. JSON-scaffolding tokens are purely structural serialization tokens, such as braces, brackets, commas, colons, and quotation marks. Non-critical tokens include schema field names and other formatting or repeated tokens that support parseability but are not themselves scored task decisions.

Let $\mathcal{C}^a$ denote the critical token positions and $\mathcal{N}^a$ denote the union of JSON-scaffolding and non-critical token positions for $a \in \{+, -\}$ after the first divergence. The remaining DPO gradient can be decomposed as

$$G_{\text{crit}} = \sum_{t \in \mathcal{C}^+} \nabla_\theta \log p_\theta\big(u_t^+ \mid x, u_{<t}^+\big) - \sum_{t \in \mathcal{C}^-} \nabla_\theta \log p_\theta\big(u_t^- \mid x, u_{<t}^-\big). \tag{A5}$$

$$G_{\text{non}} = \sum_{t \in \mathcal{N}^+} \nabla_\theta \log p_\theta\big(u_t^+ \mid x, u_{<t}^+\big) - \sum_{t \in \mathcal{N}^-} \nabla_\theta \log p_\theta\big(u_t^- \mid x, u_{<t}^-\big). \tag{A6}$$

Thus, the post-divergence DPO gradient can be summarized as

$$\nabla_\theta \mathcal{L}_{\text{DPO}} \propto -\left(G_{\text{crit}} + G_{\text{non}}\right). \tag{A7}$$

In low-separation structured preferences, the number of tokens that directly encode the scored decision is small relative to the number of serialization and non-critical tokens that remain in the completion:

$$|\mathcal{C}^+| + |\mathcal{C}^-| \ll |\mathcal{N}^+| + |\mathcal{N}^-|. \tag{A8}$$

Moreover, after the first divergence, even surface-identical JSON punctuation or schema field names may occur under different autoregressive prefixes,

so their gradient terms need not cancel. Standard DPO can therefore allocate update mass to tokens that do not determine annotation correctness. We refer to this gradient-allocation mismatch as *gradient dilution*.

This analysis also exposes a related failure mode: *preferred-token erosion*. Standard DPO optimizes only the relative sequence-level margin between the preferred and rejected completions, and therefore does not impose a lower bound on the likelihood of any individual preferred token. Let $\alpha_t^+(\theta) = \log p_\theta(u_t^+ \mid x, u_{<t}^+)$ and $\alpha_t^-(\theta) = \log p_\theta(u_t^- \mid x, u_{<t}^-)$. Expanding Eq. A2, we have

$$\Delta_\theta = \sum_{t=1}^{T^+} \left[\alpha_t^+(\theta) - \alpha_t^+(\mathrm{SFT})\right] - \sum_{t=1}^{T^-} \left[\alpha_t^-(\theta) - \alpha_t^-(\mathrm{SFT})\right]. \tag{A9}$$

This aggregate margin can increase even when some preferred-token likelihoods decrease. To see this, consider an update from θ to θ', and define $\delta_t^+ = \alpha_t^+(\theta') - \alpha_t^+(\theta)$ and $\delta_t^- = \alpha_t^-(\theta') - \alpha_t^-(\theta)$. Since p_{SFT} is fixed,

$$\Delta_{\theta'} - \Delta_\theta = \sum_{t=1}^{T^+} \delta_t^+ - \sum_{t=1}^{T^-} \delta_t^-. \tag{A10}$$

Thus, for any preferred token k with $\delta_k^+ < 0$, the margin can still increase whenever

$$\sum_{t \neq k} \delta_t^+ - \sum_{t=1}^{T^-} \delta_t^- > |\delta_k^+|. \tag{A11}$$

A successful DPO update therefore need not preserve the absolute likelihood of every correct preferred token. This is especially problematic for ontology-driven structured prediction: rare or ontology-specific schema tokens are often precisely the tokens on which the SFT model is least confident. The model may improve the relative preference margin while becoming less reliable at generating the correct structured record. Thus, a purely relative sequence-level objective provides no token-level protection for under-confident preferred schema tokens.

The diagnostics in Figure 3 of the main paper support this mechanism-level interpretation. TAB-PO's barrier activity is concentrated on critical schema tokens, with much weaker activation on JSON scaffolding and non-critical tokens. The gradient-mass heatmaps show that sequence-level DPO

variants can allocate substantial update mass to serialization or non-critical tokens, whereas TAB-PO concentrates learning signal on the sparse tokens that determine structured prediction correctness avoiding *gradient dilution*. The preference-probability and reference-adjusted-margin curves further show that TAB-PO maintains strong preference separation while applying targeted restoration to under-confident preferred tokens avoiding *token erosion*.

Appendix B. Auxiliary Loss Components for Structured Prediction

TAB-PO targets ontology-driven structured prediction, where most serialized tokens encode schema scaffolding while only a small subset determines semantic labelling, textual grounding, or relational linking correctness. In addition to the core preference objective and confidence-gated barrier, we consider three auxiliary loss components that address recurring pathologies in structured prediction datasets: sparse critical tokens, large variation in completion length, and long-tailed ontology-label distributions.

B.1. Token-Weighted Preference Aggregation

Structured prediction errors are often localized to a few schema-defining tokens: semantic-label tokens, textual-grounding tokens, relation labels and direction pointers. Standard sequence-level aggregation weights these tokens the same as JSON punctuation, field names, and repeated schema scaffolding. Motivated by loss-reweighting methods that increase the influence of informative or under-emphasized training signals [1], we introduce token weights that concentrate the preference signal on tokens that determine structured prediction correctness.

For each serialized completion Y_s, define token-index sets $\mathcal{T}_{\mathrm{SL}}(Y_s)$, $\mathcal{T}_{\mathrm{TG}}(Y_s)$, and $\mathcal{T}_{\mathrm{RL}}(Y_s)$ for semantic labelling, textual grounding, and relational linking, respectively. The remaining completion tokens are assigned to $\mathcal{T}_0(Y_s)$. We define

$$w_t(Y_s) = w_{\mathrm{SL}}\mathbb{I}[t \in \mathcal{T}_{\mathrm{SL}}(Y_s)] + w_{\mathrm{TG}}\mathbb{I}[t \in \mathcal{T}_{\mathrm{TG}}(Y_s)] + w_{\mathrm{RL}}\mathbb{I}[t \in \mathcal{T}_{\mathrm{RL}}(Y_s)] + w_0\mathbb{I}[t \in \mathcal{T}_0(Y_s)], \tag{B1}$$

where $w_{\mathrm{SL}}, w_{\mathrm{TG}}, w_{\mathrm{RL}} \geq w_0 \geq 0$. The token-weighted completion log-likelihood is

$$\mu_\theta^w(Y_s \mid x) = \sum_{t=1}^{T} w_t(Y_s) \log p_\theta(u_t \mid x, u_{<t}). \tag{B2}$$

Replacing the unweighted sequence likelihood with μ_θ^w gives the weighted reference-adjusted advantage

$$\begin{aligned}\Delta_\theta^w(x, Y_s^+, Y_s^-) = &\Big[\mu_\theta^w(Y_s^+ \mid x) - \mu_\theta^w(Y_s^- \mid x)\Big] \\ &- \Big[\mu_{\mathrm{SFT}}^w(Y_s^+ \mid x) - \mu_{\mathrm{SFT}}^w(Y_s^- \mid x)\Big],\end{aligned} \tag{B3}$$

where p_{SFT} is fixed. The corresponding preference loss is

$$\mathcal{L}_{\mathrm{pref}}^w(\theta) = -\mathbb{E}_{\mathcal{D}_{\mathrm{pref}}}\left[\log \sigma\big(\beta \Delta_\theta^w(x, Y_s^+, Y_s^-)\big)\right]. \tag{B4}$$

This weighting changes only how token evidence is aggregated; it does not alter preference-pair construction. It is intended to increase the effective learning signal on semantic labels, grounded spans, and relational pointers while reducing gradient dilution from shared serialization scaffolding.

A corpus-level diagnostic over the canonical serialized gold completions further supports this design: critical serialized tokens account for only about 20.0% of PV-Miner completion tokens and 13.4% of SciERC completion tokens on average. Thus, most completion tokens are not directly responsible for the scored structured decision, making uniform sequence-level aggregation a poor match for token-critical structured prediction.

B.2. Per-Example Length Normalization

Structured outputs can differ sharply in length. Some examples contain a single short grounded record, whereas others contain many records, long evidence spans, or multiple relation and coreference links. Length effects are a known source of bias in sequence scoring and generation [2, 3]. In structured prediction, this issue is especially prominent: without normalization, long serialized completions can contribute more total gradient mass simply because they contain more completion tokens, thereby biasing the model toward longer outputs and over-emphasizing examples with many records or long evidence spans rather than examples with more difficult structured decisions.

For instance, PV-Miner completions range from 35 to 1,140 serialized tokens, a 32.6× difference, while non-empty SciERC completions range from 116 to 3,112 serialized tokens, a 26.8× difference. This variation is driven by the number of extracted records, span length, and relation/coreference structure. To make each preference pair contribute at a comparable scale

regardless of serialized length, we normalize each weighted log-likelihood by its own token-weight mass:

$$Z_w(Y_s) = \max\left(\epsilon, \sum_{t=1}^{T} w_t(Y_s)\right), \tag{B5}$$

where $\epsilon > 0$ is a small constant for numerical stability. The length-normalized weighted completion log-likelihood is

$$\begin{aligned}\bar{\mu}_\theta^w(Y_s \mid x) = \frac{1}{Z_w(Y_s)} \sum_{t=1}^{T} w_t(Y_s) \\ \log p_\theta(u_t \mid x, u_{<t}).\end{aligned} \tag{B6}$$

When length normalization is enabled, the reference-adjusted advantage becomes

$$\begin{aligned}\bar{\Delta}_\theta^w(x, Y_s^+, Y_s^-) = \Big[\bar{\mu}_\theta^w(Y_s^+ \mid x) - \bar{\mu}_\theta^w(Y_s^- \mid x)\Big] \\ - \Big[\bar{\mu}_{\mathrm{SFT}}^w(Y_s^+ \mid x) - \bar{\mu}_{\mathrm{SFT}}^w(Y_s^- \mid x)\Big].\end{aligned} \tag{B7}$$

The length-normalized preference loss is

$$\mathcal{L}_{\mathrm{pref}}^{w,\mathrm{LN}}(\theta) = -\mathbb{E}_{\mathcal{D}_{\mathrm{pref}}}\Big[\log \sigma\Big(\beta \bar{\Delta}_\theta^w(x, Y_s^+, Y_s^-)\Big)\Big]. \tag{B8}$$

The same normalization is applied to the confidence-gated barrier. Let $g_t^\theta(x, u^+) \in \{0, 1\}$ denote the TAB-PO gate for preferred token u_t^+. The weighted, length-normalized barrier is

$$\mathcal{L}_{\mathrm{barrier}}^{w,\mathrm{LN}}(\theta) = \mathbb{E}_{\mathcal{D}_{\mathrm{pref}}}\left[\frac{\sum_{t=1}^{T^+} g_t^\theta(x, u^+) w_t(Y_s^+)\big[-\log p_\theta(u_t^+ \mid x, u_{<t}^+)\big]}{\max\Big(\epsilon, \sum_{t=1}^{T^+} g_t^\theta(x, u^+) w_t(Y_s^+)\Big)}\right]. \tag{B9}$$

When no preferred tokens are gated, the numerator is zero and the barrier contributes zero. This normalization ensures that examples with long spans, many records, or many relational links do not dominate optimization solely through completion length.

B.3. Class-Balanced Example Reweighting

Structured prediction datasets often have long-tailed ontology distributions. This is apparent in the frequency plots for PV-Miner Code and Sub-code labels (Figures B1a and B1b) and for SciERC Entity and Relation types (Figure B2). In PV-Miner, the most frequent Code, `PartnershipPatient`, appears 671 times, whereas the least frequent Code, `SharedDecisionProvider`, appears 108 times. The Sub-code distribution is more extreme: `Clinical Care` appears 731 times, while `acknowledgePatientExpertiseKnowledge` appears once. In SciERC, `OtherScientificTerm` appears 1,536 times compared with 231 occurrences of `Metric`, and `Used-for` appears 1,687 times compared with 166 occurrences of `Compare`. These skews can cause common labels to dominate minibatch updates and leave rare semantic or relational labels under-optimized.

We therefore apply example-level class-balanced reweighting using the effective-number principle [4]. Let L_{bal} denote the task-specific set of ontology labels used for class balancing. For a hierarchical labelling task, this set may include coarse and fine-grained semantic labels. For a relational extraction task, it may include entity labels, relation labels, and, when applicable, coreference or linking decisions.

For each gold structured output $Y^{(i)}$, let $\text{Lab}(Y^{(i)}) \subseteq L_{\text{bal}}$ denote the set of ontology labels selected for class balancing that appear in the labels or typed links of its records. For each such label $\ell \in L_{\text{bal}}$, let n_ℓ be its frequency in the training set. We define

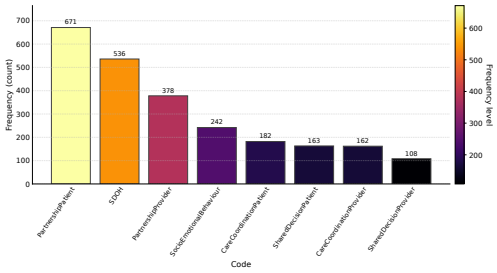
$$\eta(\ell) = \frac{1-\rho}{1-\rho^{n_\ell}}, \qquad \rho \in [0, 1). \tag{B10}$$

The example weight is

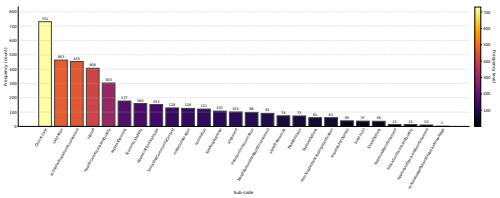
$$\begin{aligned} \omega_i &= \max_{\ell \in \text{Lab}(Y^{(i)})} \eta(\ell), \\ \omega_i &\leftarrow \min(\omega_i, \omega_{\max}). \end{aligned} \tag{B11}$$

If $\text{Lab}(Y^{(i)}) = \emptyset$, we set $\omega_i = 1$. The max operation emphasizes examples containing at least one rare ontology label, while clipping by $\omega_{\max}$ prevents unstable updates from extremely rare labels.

Let $\mathcal{L}^{(i)}_{\text{TAB-PO}}(\theta)$ denote the per-example TAB-PO loss, including the preference term and confidence-gated barrier, optionally with token weighting and

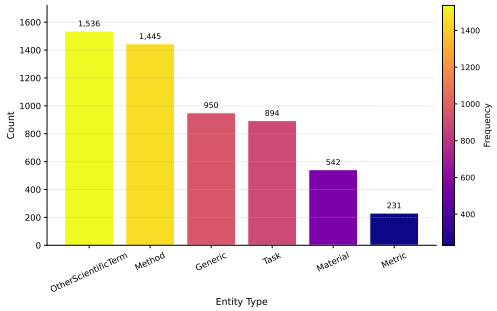


(a) Code-level PV-Miner labels. Counts reflect annotation instances rather than unique messages.

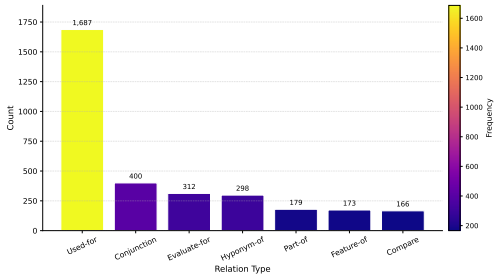


(b) Sub-code-level PV-Miner labels. Sub-codes represent fine-grained categories under the Code-level ontology.

Figure B1: Frequency distributions of PV-Miner structured prediction labels across the annotated patient messages.



(a) SciERC entity types. Counts correspond to entity annotation instances in the training split.



(b) SciERC relation types. Counts correspond to relation annotation instances in the training split.

Figure B2: Frequency distributions of SciERC structured prediction labels in the training split.

length normalization. For a minibatch of size B, the class-balanced objective is

$$\mathcal{L}_{\mathrm{CB}}(\theta) = \frac{\sum_{i=1}^{B} \omega_i \mathcal{L}_{\mathrm{TAB\text{-}PO}}^{(i)}(\theta)}{\sum_{i=1}^{B} \omega_i}. \tag{B12}$$

This normalized weighted average preserves the minibatch loss scale while increasing the influence of examples containing rare semantic labels, rare relation labels, or rare linking decisions.

B.4. Effect of the Auxiliary Components

Table B1 shows that each auxiliary component improves the metric most aligned with its intended role. Class balancing primarily strengthens ontology-

Table B1: F1 results of Plain TAB-PO and TAB-PO variants with class balancing, length normalization, and token weighting on PV-Miner and SciERC. Cells for the modified TAB-PO settings are colored by the F1 change relative to Plain TAB-PO; green indicates improvement and red indicates degradation, with stronger color intensity indicating a larger absolute change.

(a) PV-Miner

Setting	Code	Sub-code	Span
Plain TAB-PO	80.49	78.08	85.29
Plain TAB-PO + Class balancing	81.39	78.77	85.59
Plain TAB-PO + Length normalization	81.79	78.32	86.37
Plain TAB-PO + Token weighting	80.86	79.58	85.69

(b) SciERC

Setting	Entity	Relation	Span	Coreference
Plain TAB-PO	91.04	46.67	84.98	35.11
Plain TAB-PO + Class balancing	91.62	47.21	85.01	35.68
Plain TAB-PO + Length normalization	91.38	46.97	85.63	34.83
Plain TAB-PO + Token weighting	91.44	47.09	85.12	35.81

level prediction: on PV-Miner, Code F1 increases from 80.49 to 81.39 and Sub-code F1 increases from 78.08 to 78.77. Token weighting gives the clearest gain on the fine-grained Sub-code metric, increasing PV-Miner Sub-code F1 from 78.08 to 79.58, suggesting that placing more loss mass on schema-critical label tokens helps preserve fine-grained validity mappings. Length normalization most directly improves textual grounding: Span F1 increases from 85.29 to 86.37 on PV-Miner and from 84.98 to 85.63 on SciERC. Together, these trends support the intended decomposition: class balancing addresses long-tailed ontology labels, token weighting combats critical-token sparsity, and length normalization reduces optimization bias from variable-length grounded completions.

Appendix C. Prompt Engineered Instruction

This appendix provides the design rationale and failure-mode analysis for the six prompt modules (M1–M6) introduced in Section 4 of the main paper. The main text describes the template and its empirical effect; this appendix explains why each module is included and how the modules interact. The template is designed for, and can be used across, a broad class of ontology-constrained structured prediction tasks in which the model must emit schema-valid outputs containing semantic labels, grounded textual evidence, and, when required, typed relational links.

C.1. M1: Hierarchical XML Structuring

M1 organizes the instruction into semantically tagged blocks, including `<role>`, `<performance_target>`, `<task>`, and `<structured_prediction_ontology>`. The model does not parse these tags as formal XML; rather,

the tags function as lightweight structural delimiters that separate global behavioral instructions from local task-specific decision rules.

This structure serves two purposes. First, it creates explicit scope boundaries inside a long instruction. Structured prediction prompts often contain role definitions, ontology inventories, disambiguation rules, metadata controls, reasoning steps, and output schemas. Without segmentation, these constraints can interfere with one another or become less salient as the context grows. The tagged structure mitigates this by assigning each type of instruction a clear location and semantic scope. Second, the structure supports modular substitution: a new task can replace the ontology, metadata variables, grounding rules, relational-link inventory, and output schema while preserving the same instruction architecture. This improves portability across structured prediction settings and reduces format drift in long-context generation [5].

C.2. M2: Expert-Curated Disambiguation Rules

M2 encodes expert-derived decision boundaries for cases that are likely to be confused by an instruction-tuned model. These rules may distinguish confusable semantic labels, specify textual-grounding boundaries, define when a candidate Span is too broad or too narrow, or determine whether a relational link is valid. For relational tasks, the rules can additionally specify constraints over the structured relation fields, such as valid evidence pairings, relation type, link directionality, scope, and metadata-dependent validity conditions.

The purpose of this module is to operationalize the annotation manual inside the prompt. Rather than relying on the model to infer boundary conditions from label names alone, M2 states the decision criteria explicitly. This is important because structured prediction evaluation is often sensitive to small local errors: a near-correct label, an over-extended Span, an invalid relation type, an incorrect evidence pairing, or an incorrect sentence reference in a coreference record can change the scored output. Crisp disambiguation rules therefore reduce ontology-boundary, grounding-boundary, and relational-linking errors [6, 7].

C.3. M3: Reasoning Scaffold as Structured Verification

M3 provides a task-specialized reasoning scaffold. The scaffold decomposes prediction into context and metadata analysis, candidate-unit decomposition, semantic label matching, textual grounding with boundary verification, relational linking with structured-field validity verification, and final cross-validation before output emission. Unlike generic chain-of-thought prompting,

Table C1: Zero-shot F1 results under baseline (Base) and prompt-engineered instructions (Prompt) on PV-Miner and SciERC. F1 cells under Prompt are colored by the relative F1 change from baseline to prompt-engineered instruction. Stronger color intensity indicates a larger relative change.

(a) PV-Miner

Model	Code		Sub-code		Span	
	Base	Prompt	Base	Prompt	Base	Prompt
Llama-3.3-70B	60.11	**62.25**	36.13	**43.71**	55.04	**60.86**
Llama-3.1-8B	0.00	**47.09**	0.00	**20.84**	55.24	**54.15**
Qwen2.5-7B	31.71	**42.87**	22.43	**25.78**	40.29	**46.72**
Qwen2.5-14B	56.60	**60.60**	28.80	**37.74**	44.68	**47.55**
Llama-3.2-3B	22.96	**38.24**	10.54	**13.22**	35.29	**38.98**
Qwen2.5-1.5B	19.20	**22.64**	1.95	**17.85**	16.25	**21.16**

(b) SciERC

Model	Entity		Relation		Span		Coreference	
	Base	Prompt	Base	Prompt	Base	Prompt	Base	Prompt
Llama-3.3-70B	78.59	**81.08**	14.82	**16.13**	59.09	**65.10**	0.29	**1.72**
Llama-3.1-8B	73.95	**74.62**	5.64	**10.09**	53.72	**60.71**	0.00	**0.44**
Qwen2.5-7B	70.23	**78.68**	10.90	**13.33**	58.17	**66.60**	0.88	**1.01**
Qwen2.5-14B	78.83	**79.29**	12.86	**18.82**	64.44	**68.01**	5.08	**6.55**
Llama-3.2-3B	27.04	**74.16**	0.88	**3.27**	31.44	**45.84**	0.00	**0.09**
Qwen2.5-1.5B	62.21	**68.17**	4.35	**5.39**	29.98	**41.56**	0.07	**0.31**

the scaffold is not intended to elicit free-form explanation; it is a verification routine aligned with the fields of the structured output.

This module addresses reasoning shortcuts and incomplete record construction. A model may identify a plausible label without verifying the evidence Span, or it may extract valid units without checking whether the relation type, evidence, and associated sentence identifier fields are jointly valid under the task schema. M3 forces these decisions to be considered in sequence: first identify candidate units, then assign labels, then verify grounding, then validate relational-link fields, and finally check consistency across the full output. This makes the model less likely to emit records that are syntactically valid but semantically or relationally inconsistent [8].

C.4. M4: Metadata-Aware Decision Logic

M4 exposes task metadata as explicit control variables. Depending on the task, these variables may include speaker role, message direction, sentence identifier, document identifier, section type, temporal context, modality, or other task-specific controls. The key design choice is to make metadata visible to the model as a constraint rather than leaving it as implicit context.

This module narrows the valid search space for structured prediction. Metadata can determine which labels are admissible, which Spans are in scope, which sentence identifier values are valid, and which relational records are possible under the declared schema. By converting latent contextual attributes into observed control signals, M4 reduces metadata-conditioned confusion and prevents plausible but invalid predictions. Its advantage is especially clear in tasks where two records may look lexically similar but differ because of speaker role, sentence context, document section, or relational structure [9, 10].

C.5. M5: Structured Output Schema Contract

M5 specifies the output as a machine-parseable schema contract. The contract defines the required record fields, the representation of grounded evidence, semantic-label validity constraints, relational-link validity constraints, and task-specific coverage requirements. Depending on the task, the schema may specify fields for semantic labels, grounded evidence units, record identifiers, typed links between records or spans, sentence- or document-level provenance, and any additional task-specific attributes required for valid structured prediction.

The schema contract has both modeling and evaluation advantages. At inference time, it reduces format drift by giving the model a fixed structure to fill rather than an unconstrained response format. At evaluation time, it enables deterministic parsing and reproducible metric computation. M5 therefore protects the interface between model generation and automatic evaluation [7, 11].

C.6. M6: Single-Turn Self-Validation Quality Gate

M6 adds a compact validation checklist immediately before final output emission. The checklist verifies that the output is parseable under the declared schema, that semantic labels are drawn from the allowed ontology, that grounded evidence satisfies the task requirements, that relational records satisfy the required constraints, that relevant disambiguation rules have been applied, and that the final output is defensible under expert review.

This module is an intra-generation audit rather than an iterative self-refinement procedure. The model is not asked to perform multi-turn critique-and-revision; instead, it is asked to check concrete validity conditions before emitting the final answer. This distinction is important because single-pass inference is more efficient and more reproducible, and because self-correction without external feedback can be unreliable on reasoning-intensive tasks. M6 therefore functions as a pre-submission audit: it does not add new information, but it increases the likelihood that the generated structure satisfies the constraints already declared in M1–M5 [12, 13].

C.7. Interactions Between Modules

The six modules are designed to be individually interpretable but jointly effective. M1 provides the structural anchors that make the other modules easier to localize and apply. M2 supplies expert boundary conditions that are invoked during the M3 reasoning scaffold. M4 narrows the search space

before label, grounding, or relational-link decisions are made. M5 converts those decisions into a strictly parseable schema. M6 then audits the final output against the preceding constraints.

The modules therefore address complementary failure modes. M1 and M5 reduce format drift and schema violations. M2 and M4 reduce ontology ambiguity, grounding-boundary errors, and metadata-conditioned confusion. M3 reduces reasoning shortcuts by separating semantic labelling, textual grounding, and relational-link validation into explicit verification steps. M6 reduces verification omissions by requiring a final single-turn consistency check. The cross-module advantage is that the prompt does not rely on any single mechanism: structural organization, expert rules, metadata control, stepwise reasoning, schema enforcement, and self-validation work together to constrain the model toward valid structured outputs.

C.8. Prompt Engineering Improves Zero-Shot Structured Prediction

Table C1 shows that the prompt-engineered instruction substantially improves zero-shot structured prediction before any supervised or preference-based fine-tuning. On PV-Miner, the mean F1 across all model–metric cells increases from 29.85 to 39.01, corresponding to a +9.17 F1 gain and a +30.72% relative macro improvement. Prompt engineering improves 17 of 18 PV-Miner cells, with the largest average gains on ontology-level labels: +13.85 Code F1 and +9.88 Sub-code F1, compared with +3.77 Span F1.

The same trend holds on SciERC. The mean F1 increases from 30.98 to 36.71, corresponding to a +5.73 F1 gain and a +18.50% relative macro improvement. Prompt engineering improves all 24 SciERC model–metric cells. The largest average gains occur for Entity F1 (+10.86) and Span F1 (+8.50), while Relation F1 improves by +2.93 and Coreference F1 improves by +0.63. These results show that structured prompting is an important initialization step: it improves schema adherence and task decomposition, but it does not by itself solve the harder post-SFT preference-optimization problem addressed by TAB-PO.

C.9. Confusion-Matrix Evidence for Prompt-Engineered Instruction

Table C1 quantifies the performance improvements produced by the prompt-engineered instruction over the baseline instruction. Importantly, the baseline is not an uninformed prompt: it contains the essential task definition, semantic-label inventory, grounding requirements, relational-linking requirements where applicable, and input/output schema needed for structured

prediction. The gains therefore do not simply come from adding missing task information. Instead, they support the value of organizing the same task knowledge into explicit modules for ontology salience, disambiguation, metadata-aware control, stepwise verification, schema enforcement, and self-validation.

The confusion matrices in Figures C2, C3, and C4 explain where these gains arise. At the Code level, prompt engineering reduces direction-conditioned confusions between patient- and provider-oriented categories. For example, confusions between `PartnershipPatient` and `PartnershipProvider` decrease after prompt engineering. This pattern supports the role of M4, which exposes metadata and control variables as explicit constraints that narrow the valid label search space, and M6, which checks whether the final prediction is consistent with those constraints.

At the Sub-code level, prompt engineering reduces common boundary errors among semantically adjacent communication functions. The baseline shows prominent confusions such as `signoff` $\rightarrow$ `Appreciation/Gratitude` and `salutation` $\rightarrow$ `signoff`; after prompt engineering, these confusions decrease. These reductions are consistent with M2, which encodes expert-curated decision boundaries for confusable labels, and M3, which forces candidate units to be decomposed, labelled, grounded, and cross-checked before output emission.

The SciERC entity-level confusion matrices show a similar effect for scientific information extraction. Prompt engineering reduces several high-frequency off-diagonal confusions, including `OtherScientificTerm` $\rightarrow$ `Material`, `OtherScientificTerm` $\rightarrow$ `Method`, `Method` $\rightarrow$ `OtherScientific Term`, and `Task` $\rightarrow$ `Method`. These improvements suggest that the prompt modules help the model separate ontology-near entity types whose surface forms are often lexically ambiguous. They also support the combined benefit of M1, M2, M3, M5, and M6: structural segmentation improves ontology salience, expert rules sharpen local decision boundaries, the reasoning scaffold separates candidate identification from label assignment, the schema contract enforces valid structured output, and the quality gate encourages final constraint checking.

Prompt-Engineered Instruction Template for Structured Prediction

M1 | Hierarchical XML Structuring

Separates global instructions from local decision rules; improves constraint salience in long contexts.

```
<role> [Domain expert persona and behavioral prior]
</role>
<performance_target> [Precision-oriented accuracy
expectations for semantic labelling, textual grounding,
relational linking, and schema-valid structured
prediction] </performance_target>
<task> [Task definition, input/output specification,
and structured prediction ontology constraints] </task>
<structured_prediction_ontology> [Authoritative
ontology with operational definitions for semantic
labels, textual grounding units, and relational link
types/validity conditions, where applicable]
</structured_prediction_ontology>
```

M2 | Expert-Curated Disambiguation Rules

Operationalizes the annotation manual inside the prompt; resolves ontology boundary errors using expert-curated ambiguity adjudication from the annotation process.

```
<disambiguation_rules>
  [Boundary condition 1: decision criteria for
confusable semantic label pair A vs. B]
  [Boundary condition 2: textual grounding boundary,
positional, specificity, or semantic cues]
  [Boundary condition 3: relational linking validity,
source/target role, directionality, or scope
conditions]
    . . .
  [Boundary condition K: metadata-dependent or
context-dependent distinctions]
</disambiguation_rules>
```

M3 | Chain-of-Thought Reasoning Scaffold

Imposes a disciplined internal verification sequence before structured output emission.

```
<reasoning_process>
  Step 1: Context and metadata analysis
  Step 2: Candidate unit decomposition and semantic
label matching
  Step 3: Textual grounding with boundary verification
  Step 4: Relational linking with argument-validity
verification
  Step 5: Cross-validation with loop-back conditions
</reasoning_process>
```

Figure C1: General modular prompt template for ontology-constrained structured prediction with semantic labelling, textual grounding, and relational linking.

Prompt-Engineered Instruction Template for Structured Prediction

M4 | Metadata-Aware Decision Logic

```
<metadata_control>
  [Expose task metadata as explicit control variables,
e.g., speaker role, message direction, sentence
identifier, document identifier, section type, temporal
context, source/target role, modality, and
task-specific control fields]
</metadata_control>
```

Converts metadata values, which are latent contextual attributes, into observed control signals; narrows the valid search space for semantic labelling, textual grounding, and relational linking.

M5 | Structured Output Schema Contract

```
<output_format>
  [Machine-parseable hard constraints: schema for
structured predictions, grounded evidence, semantic
label validity, relational linking validity, and
task-specific coverage conditions]
</output_format>
```

Shifts the task from open-ended generation to structured generation with explicit validity conditions.

M6 | Self-Validation Quality Gate

```
<quality_gate>
  [1. Output is parseable under the declared schema]
  [2. Every semantic label is drawn from the allowed
ontology]
  [3. Every relational link satisfies source/target
argument-type constraints]
    . . .
  [K-1. All disambiguation rules have been applied]
  [K. The final output is high-confidence and
defensible under expert review]
</quality_gate>
```

Single-turn intra-generation audit; constrains the model to verify concrete failure conditions within one pass.

Figure C1: (continued)

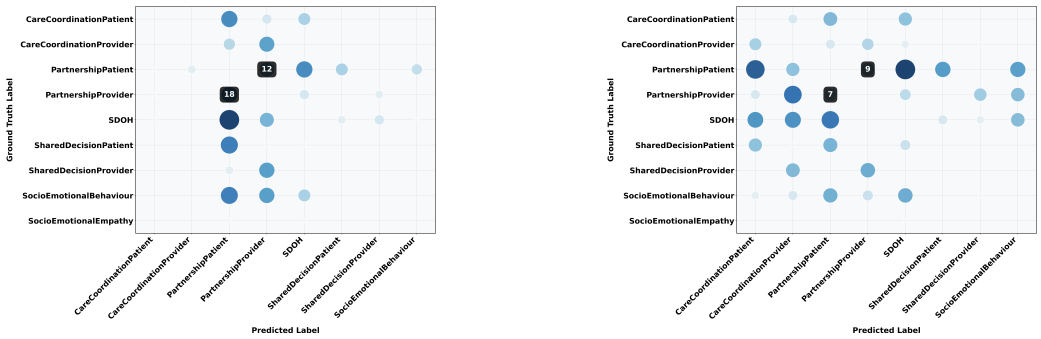


Figure C2: Code-level confusion matrices comparing baseline instruction outputs (left) and prompt-engineered instruction outputs (right) from Llama-3.2-3B-Instruct.

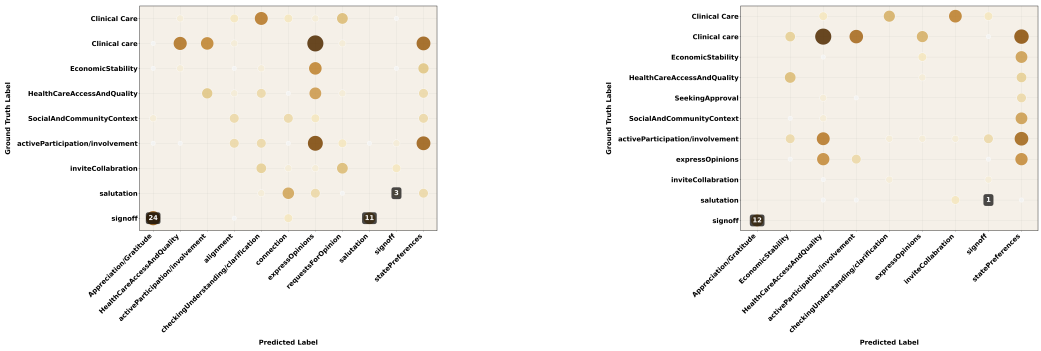


Figure C3: Sub-code-level confusion matrices comparing baseline instruction outputs (left) and prompt-engineered instruction outputs (right) from Llama-3.2-3B-Instruct. Bubble size indicates the mismatch count between ground-truth and predicted Sub-codes. Only the top 10 most frequently confused Sub-codes are shown for clarity.

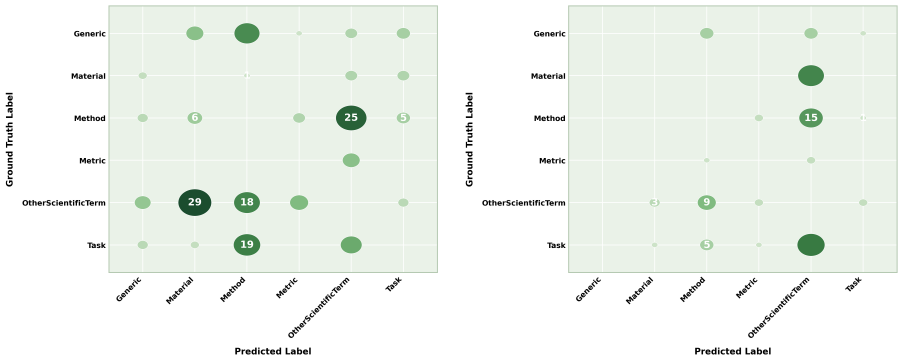


Figure C4: Entity type-level confusion matrices comparing baseline instruction outputs (left) and prompt-engineered instruction outputs (right) from Llama-3.2-3B-Instruct.

Appendix D. Task-Specific Instantiations of the Structured Prediction Formulation

This appendix instantiates the general ontology-driven structured prediction formulation introduced in Section 3 of the main paper for the two tasks studied in this work.

D.1. PV-Miner

PV-Miner requires the model to extract communication-pattern records from a patient–provider message. The objective is to predict one or more records, where each record contains a coarse-grained Code, a valid fine-grained Sub-code under that Code, and a text Span that grounds the prediction in the source message. Therefore, PV-Miner instantiates the general formulation as hierarchical semantic classification with direct textual grounding and no inter-record relations.

Let $d \in \{\mathtt{Y}, \mathtt{N}\}$ denote the message-direction metadata, where $d = \mathtt{Y}$ denotes provider→patient and $d = \mathtt{N}$ denotes patient→provider. Let L_{Code} be the set of valid Codes and $L_{\text{Sub-code}}$ be the set of valid Sub-codes. The PV-Miner ontology defines a hierarchy

$$\mathcal{H}_{\text{PV}} : L_{\text{Code}} \to 2^{L_{\text{Sub-code}}}, \tag{D1}$$

where $\mathcal{H}_{\text{PV}}(\kappa)$ gives the set of Sub-codes valid under Code κ.

The model maps

$$f_\theta : (s, d, \mathcal{O}_{\text{PV}}) \longmapsto \widehat{Y}_{\text{PV}}. \tag{D2}$$

Record Instantiation. Each PV-Miner record is instantiated as

$$y_i^{\text{PV}} = \big((\kappa_i, \varsigma_i), \{\pi_i\}, \emptyset\big), \tag{D3}$$

where

$$\kappa_i \in L_{\text{Code}}, \qquad \varsigma_i \in \mathcal{H}_{\text{PV}}(\kappa_i), \qquad \pi_i \in \mathcal{S}(s). \tag{D4}$$

Thus,

$$\widehat{Y}_{\text{PV}} = \Big\{ \big((\kappa, \varsigma), \{\pi\}, \emptyset\big) \;\Big|\; \kappa \in L_{\text{Code}}, \varsigma \in \mathcal{H}_{\text{PV}}(\kappa), \pi \in \mathcal{S}(s) \Big\}. \tag{D5}$$

Here, $\ell_i = (\kappa_i, \varsigma_i)$ is the ordered hierarchical label tuple, $\Pi_i = \{\pi_i\}$ is the grounding span, and $R_i = \emptyset$ because PV-Miner records do not point to other records.

D.2. SciERC

SciERC requires the model to extract a relational information graph from a scientific document. The objective is to predict typed entity records grounded in text, typed relation records connecting entity records, and coreference cluster records grouping entity records that refer to the same underlying scientific concept. Thus, SciERC instantiates the general formulation as a record graph with grounded entity records and structural relation/coreference records. The model maps

$$f_\theta : (s, \mathcal{O}_{\text{SciERC}}) \longmapsto \widehat{Y}_{\text{SciERC}}. \tag{D6}$$

The predicted SciERC output is partitioned into three record subsets:

$$\widehat{Y}_{\text{SciERC}} = \mathcal{E} \,\dot\cup\, \mathcal{R} \,\dot\cup\, \mathcal{C}, \tag{D7}$$

where $\mathcal{E}$ is the set of entity records, $\mathcal{R}$ is the set of relation records, and $\mathcal{C}$ is the set of coreference records. Each record still follows the same general format $y_i = (\ell_i, \Pi_i, R_i)$.

Record Instantiation. Let L_{Entity} denote the SciERC entity-type inventory, L_{Relation} denote the relation-type inventory, and $\mathcal{D}(s)$ denote the set of sentence identifiers in the source document. The predicted SciERC output is partitioned into entity, relation, and coreference records:

$$\widehat{Y}_{\text{SciERC}} = \mathcal{E} \,\dot\cup\, \mathcal{R} \,\dot\cup\, \mathcal{C}. \tag{D8}$$

Entity records are typed mentions grounded in the source text:

$$e = \big(t_e, \{\pi_e\}, \emptyset\big), \qquad t_e \in L_{\text{Entity}}, \quad \pi_e \in \mathcal{S}(s). \tag{D9}$$

Here, t_e is the entity type and π_e is the text span that grounds the entity mention.

Relation records are typed directed links from one grounded entity mention to another:

$$r = \big(\{\pi_{r1}\}, t_r, \{\pi_{r2}\}\big), t_r \in L_{\text{Relation}}, \pi_{r1}, \pi_{r2} \in \mathcal{S}(s). \tag{D10}$$

Here, t_r is the relation type, and the ordered pair of grounding sets $(\{\pi_{r1}\}, \{\pi_{r2}\})$ specifies the directed relation from the first grounded entity mention to the second grounded entity mention.

Coreference records link grounded entity mentions that refer to the same underlying scientific concept:

$$c = \big(\{\pi_{c1}\}, d_1, \{\pi_{c2}\}, d_2\big), \pi_{c1}, \pi_{c2} \in \mathcal{S}(s), d_1, d_2 \in \mathcal{I}_{\text{sent}}(s). \tag{D11}$$

Here, $\{\pi_{c1}\}$ and $\{\pi_{c2}\}$ are the grounding sets for the two linked mentions, and d_1 and d_2 are their corresponding sentence identifiers.

Appendix E. Metric Definitions

Each task adopts an evaluation strategy tailored to its prediction format and semantic structure. This appendix defines the task-specific metrics used for PV-Miner and SciERC. PV-Miner is evaluated through hierarchical semantic-label prediction and textual grounding, using Code F1, Sub-code F1, and Span F1. SciERC is evaluated through scientific information extraction, using Entity F1, Relation F1, Span F1, and Coreference F1 to measure semantic labelling, textual grounding, and relational linking.

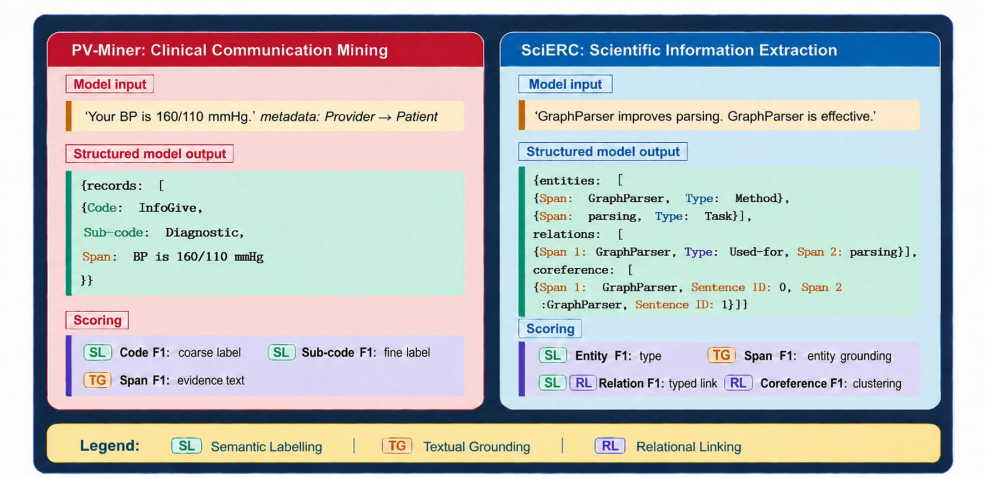


Figure E1: Examples of the two ontology-constrained structured prediction tasks evaluated in this paper. **PV-Miner** requires hierarchical clinical-communication labels with grounded evidence spans; **SciERC** requires entity typing, span grounding, relation extraction, and coreference linking.

E.1. PV-Miner

Code Classification The Code classification task is formulated as a multi-label classification problem over a predefined set of communicative Codes. Let $\widehat{\mathcal{K}}^{(i)}$ denote the predicted Code set and $\mathcal{K}^{(i)}$ the gold Code set for instance i. We compute precision, recall, and F1-score as follows:

$$\begin{aligned} \text{precision}_{\text{Code}} &= \frac{\sum_i \left|\widehat{\mathcal{K}}^{(i)} \cap \mathcal{K}^{(i)}\right|}{\sum_i \left|\widehat{\mathcal{K}}^{(i)}\right|} \\ \text{recall}_{\text{Code}} &= \frac{\sum_i \left|\widehat{\mathcal{K}}^{(i)} \cap \mathcal{K}^{(i)}\right|}{\sum_i |\mathcal{K}^{(i)}|} \\ F1_{\text{Code}} &= \frac{2 \times \text{precision}_{\text{Code}} \times \text{recall}_{\text{Code}}}{\text{precision}_{\text{Code}} + \text{recall}_{\text{Code}}}. \end{aligned} \tag{E1}$$

Sub-code Classification Sub-code classification is also evaluated as a multi-label task, where each message may be annotated with one or more Sub-codes tied to a parent Code. Let $\widehat{\mathcal{V}}^{(i)}$ and $\mathcal{V}^{(i)}$ denote predicted and gold Sub-code sets, respectively. Metrics are calculated using:

$$\begin{aligned} \text{precision}_{\text{Sub-code}} &= \frac{\sum_i \left|\widehat{\mathcal{V}}^{(i)} \cap \mathcal{V}^{(i)}\right|}{\sum_i \left|\widehat{\mathcal{V}}^{(i)}\right|} \\ \text{recall}_{\text{Sub-code}} &= \frac{\sum_i \left|\widehat{\mathcal{V}}^{(i)} \cap \mathcal{V}^{(i)}\right|}{\sum_i |\mathcal{V}^{(i)}|} \\ F1_{\text{Sub-code}} &= \frac{2\,\text{precision}_{\text{Sub-code}}\,\text{recall}_{\text{Sub-code}}}{\text{precision}_{\text{Sub-code}} + \text{recall}_{\text{Sub-code}}}. \end{aligned} \tag{E2}$$

Span For Span extraction, we evaluate each predicted evidence string against gold Spans using a **relaxed token-level matching strategy**, which combines:

- Full containment (i.e., gold Span is fully included in predicted Span or vice versa).

- Jaccard similarity (between predicted and gold Spans), with a threshold of 0.6.

Let $\widehat{\Pi}^{(i)}$ and $\Pi^{(i)}$ denote the sets of predicted and gold evidence Spans (strings) for instance i. A predicted Span $\hat{\pi} \in \widehat{\Pi}^{(i)}$ is counted as a **true positive (TP)** if it matches any gold Span $\pi \in \Pi^{(i)}$ such that $\text{Tok}(\pi) \subseteq \text{Tok}(\hat{\pi})$ or $\text{Tok}(\hat{\pi}) \subseteq \text{Tok}(\pi)$ or $\text{Jaccard}(\hat{\pi}, \pi) \geq 0.6$. Spans in $\widehat{\Pi}^{(i)}$ that fail to match any gold Span are counted as **false positives (FP)**, and Spans in $\Pi^{(i)}$ not matched by any prediction are counted as **false negatives (FN)**. Precision, recall, and F1-score are then computed as:

$$\begin{aligned} \text{precision}_{\text{Span}} &= \frac{|\text{TP}|}{|\text{TP} + \text{FP}|} \\ \text{recall}_{\text{Span}} &= \frac{|\text{TP}|}{|\text{TP} + \text{FN}|} \\ F1_{\text{Span}} &= \frac{2 \times \text{precision}_{\text{Span}} \times \text{recall}_{\text{Span}}}{\text{precision}_{\text{Span}} + \text{recall}_{\text{Span}}}. \end{aligned} \tag{E3}$$

E.2. SciERC

Entity The Entity classification task is formulated as a multi-label classification problem over a predefined set of scientific entity types. Let $\widehat{\mathcal{T}}^{(i)}_{\text{Entity}}$ denote the predicted entity-type set for instance i, and let $\mathcal{T}^{(i)}_{\text{Entity}}$ denote the corresponding gold-standard entity-type set, where these sets are obtained by projecting each entity record onto its label component t_e. We compute precision, recall, and F1-score as follows:

$$\text{precision}_{\text{Entity}} = \frac{\sum_i \left| \widehat{\mathcal{T}}^{(i)}_{\text{Entity}} \cap \mathcal{T}^{(i)}_{\text{Entity}} \right|}{\sum_i \left| \widehat{\mathcal{T}}^{(i)}_{\text{Entity}} \right|}$$

$$\text{recall}_{\text{Entity}} = \frac{\sum_i \left| \widehat{\mathcal{T}}^{(i)}_{\text{Entity}} \cap \mathcal{T}^{(i)}_{\text{Entity}} \right|}{\sum_i \left| \mathcal{T}^{(i)}_{\text{Entity}} \right|} \tag{E4}$$

$$F1_{\text{Entity}} = \frac{2\,\text{precision}_{\text{Entity}}\,\text{recall}_{\text{Entity}}}{\text{precision}_{\text{Entity}} + \text{recall}_{\text{Entity}}}.$$

Span For Entity Span extraction, we evaluate each predicted evidence string against gold Entity Spans using a **relaxed token-level matching strategy**, which combines:

- Full containment, i.e., the gold Span is fully included in the predicted Span or vice versa.
- Jaccard similarity between predicted and gold Spans, with a threshold of 0.6.

Let $\widehat{\Pi}^{(i)}_e$ and $\Pi^{(i)}_e$ denote the sets of predicted and gold evidence Spans for instance i, where each grounding set satisfies $\Pi^{(i)}_e \subseteq \mathcal{S}(s_i)$ and anchors the structured record to the source sequence s_i.

For a predicted Span $\hat{\pi_e} \in \widehat{\Pi_e}^{(i)}$ and a gold Span $\pi_e \in \Pi^{(i)}_e$, let $\text{Tok}(\cdot)$ denote the set of tokens in the Span. The relaxed Span-matching function is defined as:

$$\begin{aligned} \text{Match}_{\Pi_e}(\hat{\pi}_e, \pi_e) = &\mathbb{I}\,[\text{Tok}(\pi_e) \subseteq \text{Tok}(\hat{\pi}_e) \vee \\ &\text{Tok}(\hat{\pi}_e) \subseteq \text{Tok}(\pi_e) \ \vee \ \text{Jaccard}(\hat{\pi}_e, \pi_e) \geq 0.6] \end{aligned} \tag{E5}$$

A predicted Span $\hat{\pi} \in \widehat{\Pi}^{(i)}$ is counted as a **true positive (TP)** if it matches any gold Span $\pi \in \Pi^{(i)}$ under Match_Π. Spans in $\widehat{\Pi}^{(i)}$ that fail to match any gold Span are counted as **false positives (FP)**, and Spans in $\Pi^{(i)}$ not matched by any prediction are counted as **false negatives (FN)**. Precision, recall, and F1-score are then computed as:

$$\text{precision}_{\text{Span}} = \frac{|\text{TP}_{\text{Span}}|}{|\text{TP}_{\text{Span}}| + |\text{FP}_{\text{Span}}|}$$

$$\text{recall}_{\text{Span}} = \frac{|\text{TP}_{\text{Span}}|}{|\text{TP}_{\text{Span}}| + |\text{FN}_{\text{Span}}|} \tag{E6}$$

$$F1_{\text{Span}} = \frac{2\,\text{precision}_{\text{Span}}\,\text{recall}_{\text{Span}}}{\text{precision}_{\text{Span}} + \text{recall}_{\text{Span}}}.$$

Relation Relation extraction is evaluated by jointly checking the relation type and the ordered grounding Spans for the two relation arguments. Let a predicted relation record be represented as $\hat{r} = (\widehat{\Pi}_{r1}^{(i)}, \hat{t}, \widehat{\Pi}_{r2}^{(i)})$, where $\widehat{\Pi}_{r1}^{(i)}$ and $\widehat{\Pi}_{r2}^{(i)}$ are the predicted grounding sets for the first and second relation arguments, and $\hat{t}$ is the predicted relation type. Similarly, let a gold relation record be represented as $r = (\Pi_{r1}^{(i)}, t, \Pi_{r2}^{(i)})$.

A predicted relation is considered correct only if the relation type matches and both argument grounding sets match their corresponding gold grounding sets in the correct order:

$$\begin{aligned}\text{Match}_{\text{Relation}}(\hat{r}, r) = \mathbb{I}\Big[\hat{t} = t \\ \wedge\ \exists\, \hat{\pi}_{r1} \in \widehat{\Pi}_{r1}^{(i)}, \pi_{r1} \in \Pi_{r1}^{(i)} \text{ s.t. } \text{Match}_{\Pi}(\hat{\pi}_{r1}, \pi_{r1}) = 1 \\ \wedge\ \exists\, \hat{\pi}_{r2} \in \widehat{\Pi}_{r2}^{(i)}, \pi_{r2} \in \Pi_{r2}^{(i)} \text{ s.t. } \text{Match}_{\Pi}(\hat{\pi}_{r2}, \pi_{r2}) = 1\Big].\end{aligned} \tag{E7}$$

The ordering of relation arguments is strict: the predicted first argument grounding set $\widehat{\Pi}_{r1}^{(i)}$ must match the gold first argument grounding set $\Pi_{r1}^{(i)}$, and the predicted second argument grounding set $\widehat{\Pi}_{r2}^{(i)}$ must match the gold second argument grounding set $\Pi_{r2}^{(i)}$.

After one-to-one matching between predicted and gold relation records, precision, recall, and F1-score are computed as:

$$\text{precision}_{\text{Relation}} = \frac{|\text{TP}_{\text{Relation}}|}{|\text{TP}_{\text{Relation}}| + |\text{FP}_{\text{Relation}}|}$$

$$\text{recall}_{\text{Relation}} = \frac{|\text{TP}_{\text{Relation}}|}{|\text{TP}_{\text{Relation}}| + |\text{FN}_{\text{Relation}}|} \tag{E8}$$

$$F1_{\text{Relation}} = \frac{2\,\text{precision}_{\text{Relation}}\,\text{recall}_{\text{Relation}}}{\text{precision}_{\text{Relation}} + \text{recall}_{\text{Relation}}}.$$

Coreference Coreference is evaluated at the coreference-link level. Let a predicted coreference link be represented as $\hat{c} = (\widehat{\Pi}_{c1}^{(i)}, \hat{d}_1, \widehat{\Pi}_{c2}^{(i)}, \hat{d}_2)$, where $\widehat{\Pi}_{c1}^{(i)}$ and $\widehat{\Pi}_{c2}^{(i)}$ are the predicted grounding sets for the two linked mentions, and $\hat{d}_1$ and $\hat{d}_2$ are their predicted sentence identifiers. Similarly, let a gold coreference link be represented as $c = (\Pi_{c1}^{(i)}, d_1, \Pi_{c2}^{(i)}, d_2)$.

A predicted coreference link is considered correct only if both mention grounding sets match under relaxed Span matching and both sentence identifiers match exactly:

$$\begin{aligned}\text{Match}_{\text{Coref}}(\hat{c}, c) = \mathbb{I}\Big[& \\ & \exists \hat{\pi}_{c1} \in \widehat{\Pi}_{c1}^{(i)},\, \pi_{c1} \in \Pi_{c1}^{(i)} : \text{Match}_{\Pi}(\hat{\pi}_{c1}, \pi_{c1}) = 1 \;\wedge\; \hat{d}_1 = d_1 \\ & \wedge\; \exists \hat{\pi}_{c2} \in \widehat{\Pi}_{c2}^{(i)},\, \pi_{c2} \in \Pi_{c2}^{(i)} : \text{Match}_{\Pi}(\hat{\pi}_{c2}, \pi_{c2}) = 1 \;\wedge\; \hat{d}_2 = d_2\Big].\end{aligned} \tag{E9}$$

Thus, coreference evaluation requires relaxed textual agreement for both linked mention grounding sets and strict agreement for their sentence identifiers.

After one-to-one matching between predicted and gold coreference links, precision, recall, and F1-score are computed as:

$$\text{precision}_{\text{Coreference}} = \frac{|\text{TP}_{\text{Coreference}}|}{|\text{TP}_{\text{Coreference}}| + |\text{FP}_{\text{Coreference}}|}$$

$$\text{recall}_{\text{Coreference}} = \frac{|\text{TP}_{\text{Coreference}}|}{|\text{TP}_{\text{Coreference}}| + |\text{FN}_{\text{Coreference}}|} \tag{E10}$$

$$F1_{\text{Coreference}} = \frac{2\,\text{precision}_{\text{Coreference}}\,\text{recall}_{\text{Coreference}}}{\text{precision}_{\text{Coreference}} + \text{recall}_{\text{Coreference}}}.$$

Appendix F. Extended Related Work

This appendix extends Section 2 of the main paper by comparing the representations, preference data, and likelihood-control mechanisms most relevant to TAB-PO. The main distinction is between selecting tokens to modify a contrastive preference score and selecting preferred tokens for a separate supervised restoration loss.

Structured representations, guidelines, and admissibility. Joint scientific extraction models connect entity, relation, and coreference decisions [14]. UIE expresses heterogeneous extraction tasks through a common structured language and schema instructor [15], while GenIE constrains autoregressive extraction to knowledge-base entities and relations [16]. These are precedents for structured generation; TAB-PO's formulation organizes the semantic, grounding, and linking dimensions needed for preference construction and diagnostics. Guideline Learning derives and retrieves instructions from extraction errors [6], and GoLLIE trains models to follow annotation guidelines [7]. G&O separates content generation from output organization [11], whereas grammar-constrained decoding restricts admissible clinical extraction structures [17]. Representational and decoding constraints complement preference learning: a schema-admissible label or source-grounded span can still express an incorrect interpretation. TAB-PO contrasts such structured alternatives after supervised initialization.

Preference alignment for information extraction. ADELIE is a direct antecedent for post-SFT preference learning in information extraction [18]. It combines information-extraction instruction tuning with DPO and constructs preferences both between gold and generated answers and between sampled answers ranked by BLEU. Thus, applying preference optimization to extraction is established. TAB-PO specializes this setting through the combination of ontology-confusion-based alternatives and a preferred-token restoration term. Its rejected records encode particular task errors under declared validity rules, while its additional loss is activated by the current probabilities of preferred tokens. This comparison concerns the preference-construction protocol and objective; it does not imply that standard DPO or model-generated negatives are uniformly ineffective for extraction.

Offline preference objectives and aggregate likelihood control. RLHF learns a reward model and subsequently optimizes a policy using sampled responses

[10]. DPO replaces these stages with an offline loss on preferred and rejected reference-adjusted completion log-likelihoods [19]. IPO, derived within the ΨPO framework, targets a regularized preference margin through a squared objective [20]. SimPO removes the reference model, uses average token log-likelihood, and introduces a target reward margin [21]; Cal-DPO constrains implicit reward magnitudes [22]. These mechanisms control different properties of preference learning. DPOP is especially relevant because it studies failures on low-edit-distance pairs and adds a penalty when preferred completion likelihood falls below its reference value [23]. Ren and Sutherland analyze a squeezing effect associated with preferred-likelihood reductions [24], while Razin et al. connect likelihood displacement to similarity between preferred and rejected representations [25]. Low token edit distance motivates examining these risks in structured prediction, but it is not equivalent to the representation-similarity condition in that analysis. Moreover, an aggregate preferred-likelihood increase can coexist with a decrease at an individual correct token. A completion-level penalty can consequently remain inactive while such a local decrease occurs. TAB-PO supplies an additional loss at preferred positions selected by their current confidence.

Supervised regularization and selective restoration. SLiC-HF combines sequence-likelihood calibration with supervised regularization [26]. CPO adds preferred-output negative log-likelihood to a reference-free contrastive objective [27], and APO controls likelihood changes according to the relationship between the current model and preference data [28]. TAB-PO's contribution is therefore not supervised regularization in isolation. Its barrier uses a detached confidence gate and normalizes by the number of selected preferred tokens (Eq. 18 of the main paper). For a completion with T^+ tokens, a uniformly averaged NLL term with coefficient λ assigns coefficient λ/T^+ to each token's NLL contribution. If TAB-PO selects $K > 0$ positions, the corresponding coefficient is λ/K at selected positions and zero elsewhere. This changes the support and normalization of the additional loss. It does not imply that confident serialization tokens necessarily dominate an ungated NLL gradient, because cross-entropy gradients also depend on prediction probabilities. The gate allocates direct supervised restoration to currently low-confidence preferred positions and becomes inactive when none are selected. The threshold is an activation criterion, not a hard probability floor; shared-parameter updates need not increase every selected probability monotonically.

Token-level credit assignment and confidence selection. TDPO introduces token-wise forward-KL control [29]; TI-DPO combines gradient-attribution-based importance with a triplet-style objective [30]; and TIS-DPO estimates importance through prediction-probability differences between contrastive models [31]. T-REG derives token-level rewards through opposite prompting and uses them to regularize sequence-level preference learning [32]. ConfPO and BMC provide particularly close comparisons. ConfPO selects low-confidence tokens from both responses for preference scoring, using response-specific average-probability thresholds in its primary formulation; it also studies fixed thresholds and application to DPO [33]. BMC identifies differing tokens in correlated pairs and uses detached, confidence-dependent weights inside the reference-adjusted preference score [34]. TAB-PO preserves its core sequence-level preference score and uses confidence to select preferred positions for a separate NLL term outside the log-sigmoid. The distinction is the role and asymmetric application of restoration, rather than policy confidence or token selection alone. The core gate does not explicitly test ontology membership: concentration on schema-critical positions is an empirical diagnostic (Section 7.1 of the main paper). Explicit task-based weighting, length normalization, and class balancing are separate auxiliary components (Appendix B).

Contrastive data and ontology-informed perturbations. CLAIR creates minimally contrastive response revisions [28], and BMC constructs a pseudo-winning response by revising a losing response with the winning response as a reference [34]. DPOP also uses local corruptions in its reasoning experiments [23]. These works establish the value of controlled response differences. TAB-PO uses empirical SFT confusions and expert-observed ambiguities to guide substitutions and other structured perturbations (Section 6 of the main paper; Appendix G). Ontology and schema rules constrain admissibility, while gold annotations specify the preferred interpretation. Label and link substitutions can produce very small serialized differences; record insertion or omission can alter many tokens. The common principle is localization of the structured error, rather than uniformly small token edit distance across every perturbation family. Coupling these task-informed alternatives with confidence-gated restoration makes preference construction and optimization address the same structured decisions.

Autoregressive overlap and direct restoration. In the unweighted DPO margin, identical tokens under identical prefixes cancel before the first completion

divergence. After divergence, identical field names or punctuation can have different conditional probabilities and gradient contributions because their prefixes differ. Large surface overlap alone therefore does not establish gradient dilution. Appendix A decomposes the post-divergence signal, and Section 7.1 of the main paper examines its allocation empirically. Cancellation concerns a token's direct contribution to the margin: its probability can still change indirectly through updates to shared parameters. TAB-PO's separate restoration term can provide direct supervision at selected preferred positions even when their contrastive contributions cancel. It also addresses a distinct limitation of relative preference separation: an improved completion margin does not ensure preservation of every correct preferred-token likelihood.

Verifier-based reinforcement learning and training scope. GRPO-style optimization and reinforcement learning with verifiable rewards use sampled outputs and task-based reward signals [35, 36]. Gold structured records can likewise support programmatic rewards, including record-level F1 and task-specific correctness checks. TAB-PO addresses an offline post-SFT setting in which fixed preferred and rejected structured outputs are prepared before optimization. Its preference-training stage requires neither online policy rollouts nor a learned reward model, although data preparation still entails SFT inference, perturbation generation, validity checks, and expert review. The comparison with offline DPO-family objectives evaluates alternatives within this preference-learning setting. Verifier-based RL is complementary, with different sampling and supervision requirements. TAB-PO's focus is the joint design of task-informed structured preferences and selective preferred-token restoration.

Appendix G. Targeted TAB-PO Negative Construction

This appendix illustrates how TAB-PO constructs targeted negative outputs for ontology-driven structured prediction. Given an input x and gold structured output Y^+, TAB-PO constructs a rejected output Y^- by introducing a small, schema-valid perturbation that reflects a realistic structured prediction error. The goal is not to create arbitrary corrupted outputs, but to produce low-separation hard negatives: outputs that remain parseable and ontology-valid while differing from the gold output in the semantic label, textual grounding, or relational-linking decision that determines correctness. Figures G1, G2, and G3 below illustrate representative perturbations for PV-Miner and SciERC.

Targeted TAB-PO negative construction for PV-Miner

Legend: Semantic labelling confusion Missing record perturbation Extra record perturbation

Metadata: TO_PAT_YN=Y *(Provider→Patient)*

Input message: *No worries! Thank you for letting me know. Best wishes, Dr. Person1*

Ground truth:

```
[
  {Code: PartnershipProvider, Sub-code: connection, Span: "No worries!"},
  {Code: PartnershipProvider, Sub-code: Appreciation/Gratitude,
                              Span: "Thank you for letting me know"},
  {Code: PartnershipProvider, Sub-code: signoff, Span: "Best wishes"}
]
```

1) Semantic labelling confusion *A semantic label is perturbed while the grounded span is retained. Code–Sub-code mapping is preserved under Semantic labelling confusion.*

```
[
  {Code: PartnershipPatient, Sub-code: connection, Span: "No worries!"},
  {Code: PartnershipProvider, Sub-code: signoff,
                              Span: "Thank you for letting me know"},
  {Code: PartnershipProvider, Sub-code: signoff, Span: "Best wishes"}
]
```

2) Missing record perturbation *A valid ground-truth record is removed from the structured output.*

```
[
  {Code: PartnershipProvider, Sub-code: connection, Span: "No worries!"},
  {Code: PartnershipProvider, Sub-code: Appreciation/Gratitude,
                              Span: "Thank you for letting me know"}
]
```

Removed: `{Code: PartnershipProvider, Sub-code: signoff, Span: "Best wishes"}`

3) Extra record perturbation *An additional validly formatted but unsupported record is inserted. The Span for the extra record is chosen from the input message.*

```
[
  {Code: PartnershipProvider, Sub-code: connection, Span: "No worries!"},
  {Code: PartnershipProvider, Sub-code: Appreciation/Gratitude,
                              Span: "Thank you for letting me know"},
  {Code: PartnershipProvider, Sub-code: signoff, Span: "Best wishes"},
  {Code: SDOH, Sub-code: SocialAndCommunityContext, Span: "No worries"}
]
```

Figure G1: PV-Miner example of targeted TAB-PO negative construction.

Targeted TAB-PO negative construction for SciERC

Legend: Semantic labelling confusion Relational linking confusion Missing record perturbation Extra record perturbation

Input sample: *GraphParser improves parsing. GraphParser is effective.*

Ground-truth:

```
{
  entities:    [{Span: GraphParser, Type: Method}, {Span: parsing, Type: Task}],
  relations:   [{Span 1: GraphParser, Type: Used-for, Span 2: parsing}],
  coreference: [{Span 1: GraphParser, Sentence ID: 0,
                 Span 2: GraphParser, Sentence ID: 1}]
}
```

1) Semantic labelling confusion *A semantic label is perturbed while the grounded span and structured record are retained. For SciERC, this can perturb an entity type, or relation type.*

```
{
  entities:    [{Span: GraphParser, Type: Method}, {Span: parsing, Type: Method}],
  relations:   [{Span 1: GraphParser, Type: Evaluate-for, Span 2: parsing}],
  coreference: [{Span 1: GraphParser, Sentence ID: 0,
                 Span 2: GraphParser, Sentence ID: 1}]
}
```

2) Relational linking confusion *A relational link is perturbed while the spans remain grounded in the input. This may involve reversing Span 1 and Span 2, confusing one linked span with another entity span, or modifying a coreference link by substituting another entity span with a valid sentence ID.*

```
{
  entities:    [{Span: GraphParser, Type: Method}, {Span: parsing, Type: Task}],
  relations:   [{Span 1: parsing, Type: Used-for, Span 2: GraphParser}],
  coreference: [{Span 1: parsing, Sentence ID: 0,
                 Span 2: GraphParser, Sentence ID: 1}]
}
```

Figure G2: SciERC semantic-label and relational-link confusion examples for TAB-PO negative construction.

3) Missing record perturbation *A valid ground-truth record is removed from the structured output. For SciERC, this may involve removing an entity, relation, or coreference link; if an entity is removed, its associated relations and coreference links are also removed to maintain structural validity.*

```
{
  entities:    [{Span: GraphParser, Type: Method}],
  relations:   [{Span 1: GraphParser, Type: Used-for, Span 2: parsing}],
  coreference: [{Span 1: GraphParser, Sentence ID: 0,
                 Span 2: GraphParser, Sentence ID: 1}]
}
```

Removed: {Span: parsing, Type: Task}

4) Extra record perturbation *An additional validly formatted but unsupported record is inserted. For entity-level extra records, the span is chosen from the input text.*

```
{
  entities:    [{Span: GraphParser, Type: Method}, {Span: parsing, Type: Task},
                {Span: effective, Type: Metric}],
  relations:   [{Span 1: GraphParser, Type: Used-for, Span 2: parsing}],
  coreference: [{Span 1: GraphParser, Sentence ID: 0,
                 Span 2: GraphParser, Sentence ID: 1}]
}
```

Figure G3: SciERC missing-record and extra-record perturbation examples for TAB-PO negative construction.

PV-Miner focuses on hierarchical clinical annotation, where negative construction targets Code/Sub-code confusions, missing grounded records, and unsupported extra records. SciERC extends the same idea to scientific relational extraction, where perturbations can affect entity labels, relation labels, relation arguments, coreference links, missing records, and extra records. Across all perturbation types, the rejected output remains close to the preferred output while changing the specific ontology-bound decision that determines correctness. TAB-PO is designed to overcome the disadvantage of low-separation structured preferences: it converts preference optimization into a targeted signal for schema-critical tokens rather than allowing the update to be diluted across shared serialization structure.

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